

Design and Computational Optimization of a High-Q Silicon Nitride Microring Resonator for Purcell Enhancement of Diamond NV Center Emission

EE798I Course Project
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Abstract

This report presents the computational design and optimization of a silicon nitride (SiN) on silica (SiO₂) microring resonator (MRR) for enhancing the spin-photon interface of a diamond nitrogen-vacancy (NV) center. The objective is to leverage the Purcell effect to increase the spontaneous emission rate into the zero-phonon line (ZPL) at a wavelength of $\lambda \approx 637$ nm. A 2D axisymmetric model was simulated using an eigenfrequency analysis in COMSOL Multiphysics, an approach that provides a computationally efficient pathway to model the whispering gallery modes (WGMs) of a 3D resonator. Through a systematic optimization involving mesh convergence and ring radius sweeps, an optimal design was determined for the 354th-order WGM. The simulated resonator, with a radius of approximately $R = 20$ μm and a $250 \text{ nm} \times 500 \text{ nm}$ waveguide cross-section, exhibits a high radiation-limited Quality Factor (Q) of 167,058. Post-processing analysis yields an effective mode volume (V_{eff}) of $1.033 \times 10^{-16} \text{ m}^3$. These parameters result in a calculated Purcell factor (F_p) of approximately 3.86. This analysis highlights the fundamental trade-off in microring geometries between achieving a high Q-factor (which demands a large radius) and a small V_{eff} . The modest F_p suggests that while the MRR platform is easily fabricated, geometries with 3D confinement, such as photonic crystal cavities, are required to achieve the Q/V_{eff} ratios necessary for a high-efficiency quantum interface.

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1 Introduction

The diamond nitrogen-vacancy (NV) center has emerged as a premier platform for solid-state quantum information and sensing, offering long spin coherence times even at room temperature [1, 2]. A critical challenge in scaling this platform for quantum networks lies in its inefficient spin-photon interface [3]. This inefficiency is twofold: first, diamond’s high refractive index ($n \approx 2.4$) traps most emitted photons via total internal reflection (TIR), and second, the coherent Zero-Phonon Line (ZPL) emission at 637 nm accounts for only $\sim 3 - 4\%$ of the total emission, with the rest lost to a broad, incoherent phonon sideband (PSB).

A solution lies in cavity quantum electrodynamics (cQED). By placing the NV center in a high-Quality-factor (Q) and low-mode-volume (V_{eff}) optical microcavity resonant with the ZPL, the spontaneous emission rate into this coherent channel can be selectively enhanced. This phenomenon, the Purcell effect [5], can “funnel” emission from the PSB into the ZPL, dramatically increasing the branching ratio and transforming the NV center into an efficient single-photon source.

Fabricating high-Q cavities directly in diamond is extremely difficult [3]. A more scalable “hybrid” approach involves coupling nanodiamonds to a photonic integrated circuit (PIC) fabricated using mature CMOS-compatible processes. As silicon is opaque at 637 nm, silicon nitride (SiN) has become the material of choice. It offers a wide transparency window, a high refractive index ($n \approx 2.02$), and can be deposited on a low-index silicon dioxide (SiO₂) substrate ($n \approx 1.45$), enabling high-Q resonators. Nitrogen-rich SiN, in particular, mitigates the autofluorescence that plagues other variants at visible wavelengths [6].

The objective of this work is the computational design and optimization of a SiN microring resonator (MRR) to serve as this hybrid cavity. Using an eigenfrequency analysis in COMSOL Multiphysics, this report details the theoretical basis, simulation methodology, and optimization procedure used to design a high-Q MRR resonant at 637 nm. We extract the Q and V_{eff} of the resonant mode, calculate the resultant Purcell factor, and critically assess the viability of the MRR geometry for this quantum application by comparison with state-of-the-art literature.

2 Theoretical Foundations

2.1 The Diamond NV⁻ Center Electronic Structure

The NV center is a point defect in Diamond.[3] We focus on the negatively charged NV⁻ state, which has a total spin $S = 1$.

2.1.1 Spin-Triplet Manifolds

The NV⁻ electronic structure is defined by a spin-triplet ground state and a spin-triplet excited state.[6]

- **Ground State:** Due to spin-spin interactions, the ground state exhibits a zero-field splitting between the $m_s = 0$ (magnetic sub-level) and the degenerate $m_s = \pm 1$ sub-levels.[6] These long-lived spin sub-levels form the fundamental qubit.
- **Excited State (3E):** The excited state has a similar but smaller, ZFS.

The optical transition connecting the ground state and the excited state is the source of the 637 nm ZPL emission.[7]

2.1.2 Spin Initialization and Readout Mechanism

The utility of the NV center as a qubit stems from its unique spin-selective optical properties.[1, 6]

1. **Optical Pumping (Initialization):** When the NV center is illuminated with an off-resonant green laser (e.g., 532 nm), it is excited from the ground state to the excited state.
2. **Spin-Selective Intersystem Crossing (ISC):** From the excited state, the system has two competing decay paths:
 - **Radiative Decay:** It can decay radiatively back to the ground state, emitting a photon (either ZPL or PSB). This path is spin-conserving.
 - **Non-Radiative ISC:** The $m_s = \pm 1$ excited states have a high probability of undergoing a non-radiative intersystem crossing to the singlet manifold. From there, the system decays non-radiatively back to the $m_s = 0$ ground state.[1] The $m_s = 0$ state has a much lower probability of this ISC.

This process has two profound consequences. First, after a few optical cycles, the entire population is "pumped" into the $m_s = 0$ ground state, providing a robust spin initialization. Second, it enables spin readout: the $m_s = 0$ state is "bright" (high fluorescence) as it cycles radiatively, while the $m_s = \pm 1$ states are "dark" as they are rapidly shelved in the non-emissive singlet pathway.

A critical design consideration emerges from this mechanism. The spin readout relies on the competition between the radiative decay rate (Γ_{rad}) and the ISC rate (Γ_{ISC}).[1] If the Purcell factor is so large that the new radiative rate ($F_p \cdot \Gamma_{rad}$) becomes significantly faster than Γ_{ISC} , the $m_s = \pm 1$ states will decay radiatively before they can cross to the singlet state. This would make all spin states "bright", destroying the fluorescence contrast and rendering optical spin readout impossible. Therefore, the goal is not a maximal F_p , but an optimal F_p that maximizes ZPL emission while preserving the readout signal-to-noise ratio.

2.2 The Purcell Effect

The Purcell effect describes the enhancement of spontaneous emission by a cavity. The spontaneous emission rate (Γ) is governed by Fermi's Golden Rule and is proportional to the Local

Density of States (LDOS) [5]. In free space, the LDOS is a smooth continuum, yielding a constant decay rate Γ_0 . A cavity reshapes the LDOS, concentrating it into discrete, high-amplitude resonant modes.

By placing an emitter (like an NV center) in a cavity, we can selectively enhance its emission rate. The Purcell factor (F_p) is the ratio of the enhanced decay rate (Γ_c) to the free-space rate (Γ_0). Assuming the emitter is at the cavity's field maximum ($\mathbf{r}_{max}$), spectrally resonant ($\omega = \omega_c$), and its dipole is aligned with the field, the F_p is given by the canonical formula [1, ?]:

$$F_p = \frac{3}{4\pi^2} \left(\frac{\lambda}{n} \right)^3 \left(\frac{Q}{V_{eff}} \right)$$

Here, n is the refractive index of the host material, λ is the free-space wavelength, Q is the cavity Quality Factor, and V_{eff} is the effective mode volume. This equation forms the central link between the project's physical goal (maximizing F_p) and the computational task (simulating Q and V_{eff}).

2.3 Photonic Microring Resonator (MRR) Theory

The chosen cavity, a microring resonator, confines light in a circular waveguide via Total Internal Reflection (TIR), supporting so-called Whispering Gallery Modes (WGMs).

2.3.1 Whispering Gallery Modes (WGMs)

Light guidance in the curved waveguide is analogous to the acoustic "whispering gallery" phenomenon. A wave traveling within the high-index core (SiN) strikes the curved outer boundary at a shallow angle. If this angle is greater than the critical angle for TIR, the light is perfectly reflected and remains confined, propagating around the ring's circumference.

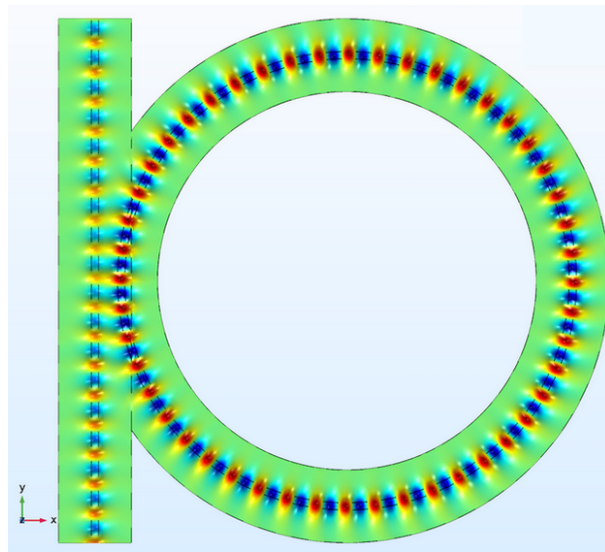


Figure 1: Whispering Gallery Modes of a Micro Ring Resonator

2.3.2 Resonance Condition

A WGM is a traveling wave. For this wave to resonate, it must constructively interfere with itself on each round trip. This condition is met when the total phase shift acquired in one circumference ($L = 2\pi R$) is an integer multiple (m) of 2π .

$$\phi_{\text{round-trip}} = m \cdot 2\pi$$

The phase shift per unit length is the propagation constant β . Thus, $\phi_{\text{round-trip}} = \beta L$. The propagation constant is related to the *effective refractive index*, n_{eff} , of the guided mode: $\beta = k_0 n_{\text{eff}} = \frac{2\pi}{\lambda} n_{\text{eff}}$. Combining these gives the fundamental resonance condition for an MRR:

$$\left(\frac{2\pi n_{\text{eff}}}{\lambda} \right) (2\pi R) = m \cdot 2\pi \implies \mathbf{m}\lambda = \mathbf{2\pi R n_{\text{eff}}}$$

This equation justifies the initial parameter estimation, linking the mode order ($m = 354$), target wavelength ($\lambda = 637$ nm), and ring radius ($R \approx 20$ μm).

2.4 Computational Method: Eigenfrequency Analysis in COMSOL

The simulation is performed using the COMSOL Multiphysics Wave Optics Module. This module solves the time-harmonic, source-free vector wave equation, which is derived from Maxwell's equations:

$$\nabla \times (\mu_r^{-1} \nabla \times \mathbf{E}) - k_0^2 (\epsilon_r - \frac{j\sigma}{\omega\epsilon_0}) \mathbf{E} = 0$$

An Eigenfrequency study solves this equation as an eigenvalue problem. In a source-free problem, non-trivial solutions (eigenmodes) only exist for a discrete set of complex "eigenvalues" (λ). For a time-harmonic assumption $u(t) = \hat{u}e^{-\lambda t}$, the solver finds λ . For an open, lossy system, the eigenvalue is the complex angular frequency:

$$\lambda = -j\omega + \delta$$

where ω is the angular frequency and δ is the decay rate. The solver returns the complex eigenfrequency $f = \omega/2\pi = f' + if''$.

3 Computational Model and Design Optimization

This section details the implementation and step-by-step optimization of the microring resonator in the COMSOL environment.

3.1 Material and Geometry Justification

- **Core Material: Nitrogen-Rich Si₃N₄:** The core refractive index is set to $n_{\text{core}} = 2.02$ at 637 nm. The choice of N-rich SiN is critical. Standard stoichiometric SiN exhibits

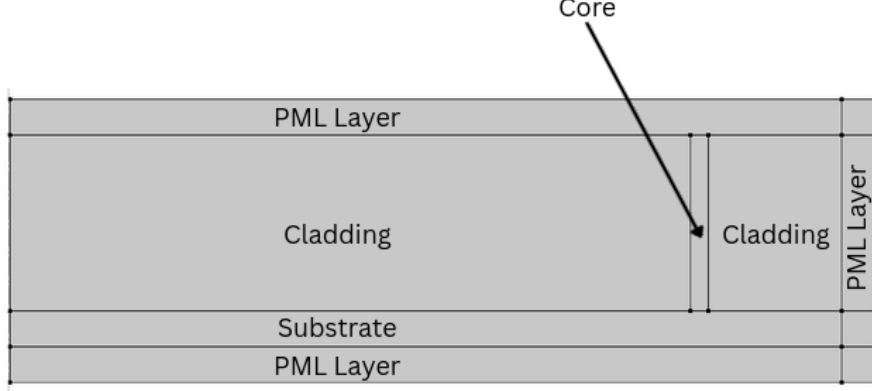


Figure 2: 2D Geometry for the Microring Resonator Design

significant autofluorescence when pumped with visible light. This background noise would obscure the single-photon ZPL signal. As demonstrated in literature, using nitrogen-rich SiN or specific annealing processes can mitigate or eliminate this background fluorescence, making the platform viable for quantum applications at visible wavelengths.

- **Cladding Material: Silicon Dioxide (SiO₂):** The substrate and overcladding are modeled as SiO₂ with $n_{clad} = 1.45$ at 637 nm. This provides a high index contrast ($\Delta n \approx 0.57$) for strong vertical confinement of the mode.
- **Computational Model: 2D Axisymmetric:** A full 3D simulation of a high- Q resonator (requiring a mesh size far smaller than the 637 nm wavelength over a ~ 125 μm circumference) is computationally prohibitive. We employ a 2D Axisymmetric model, a standard and effective approximation. This model revolves a 2D cross-section (the 250x500 nm rectangle) around the z -axis. The solver is instructed to find the eigenfrequency for a specific azimuthal mode number, $m = 354$. This reduces the problem from 3D to 2D, enabling the high-resolution sweeps required for optimization.
- **Waveguide Dimensions: 250 nm (H) $\times$ 500 nm (W):** This cross-section is chosen to ensure single-mode operation for the $\lambda = 637$ nm light. A larger cross-section could support higher-order transverse modes, which would introduce parasitic resonances and complicate the spectrum.

3.2 Initial Parameter Estimation

The resonance condition $m\lambda = 2\pi R n_{eff}$ must be met. The goal is to center the $m = 354$ mode at $\lambda = 637$ nm. The effective index n_{eff} for a 250x500 nm SiN waveguide is estimated to be $n_{eff} \approx 1.80$ (between the core $n = 2.02$ and cladding $n = 1.45$). A quick calculation provides the starting point for the radius:

$$R = \frac{m\lambda}{2\pi n_{eff}} = \frac{354 \cdot (637 \times 10^{-9} \text{ m})}{2\pi \cdot 1.80} \approx 19.98 \text{ } \mu\text{m}$$

This confirms that the initial parameter $R = 20$ μm is the correct value to target in the optimization sweep.

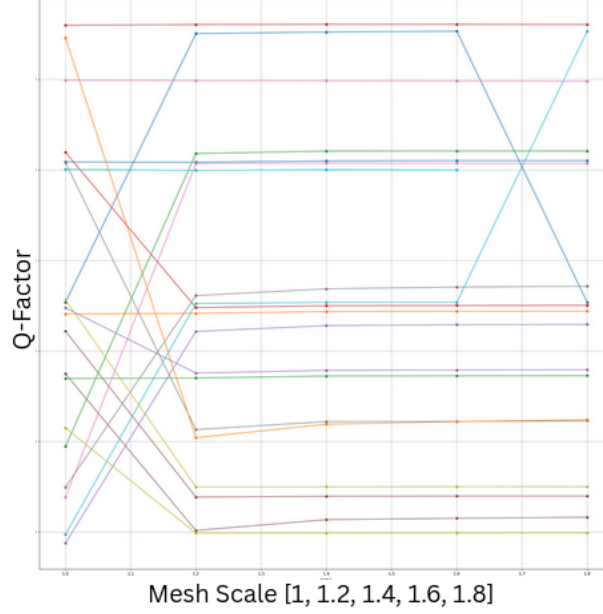


Figure 3: Q Factor v/s mesh scale plots for different eigenfrequencies for determining mesh convergence

3.3 Simulation and Optimization Procedure

The simulation was performed as an iterative, two-experiment process.

1. **Model Setup:** A 2D Axisymmetric model was created. A $250 \text{ nm} \times 500 \text{ nm}$ SiN rectangle (core) was placed at $R \approx 20 \text{ }\mu\text{m}$ from the axis of revolution. This core was embedded in a large SiO_2 domain (cladding).
2. **Boundary Conditions:** Perfectly Matched Layers (PMLs) were applied to the outer boundaries of the SiO_2 domain. This is the crucial step that simulates an open, radiating system and defines the "loss" mechanism (radiation) for the Q -factor calculation.
3. **Experiment 1: Mesh Convergence Sweep:** The Finite Element Method (FEM) divides the geometry into a discrete mesh. The accuracy of the solution is highly dependent on the mesh density. An eigenfrequency study was conducted while the mesh scale (from coarse to extremely fine) was swept. The Q -factor and resonant frequency were monitored. The "correct" mesh scale was determined when the Q -factor converged, i.e., it stopped changing significantly with further mesh refinement. This ensures the final result is independent of the computational grid.
4. **Experiment 2: Radius Optimization Sweep:** With a converged mesh, the next step was to find the exact radius R that centers the $m = 354$ mode precisely at $\lambda = 637 \text{ nm}$. A parametric sweep of the ring radius R (from $\sim 19.5 \text{ }\mu\text{m}$ to $\sim 20.5 \text{ }\mu\text{m}$) was performed within the Eigenfrequency study. This sweep tunes the resonant frequency. The optimization goal was to find the specific radius R that yielded a real part of the eigenfrequency $\text{Re}(f) = c/(637 \text{ nm})$ and simultaneously maximized the Q -factor (as Q_{rad} increases with R).

4 Results and Discussion

The optimization procedure yielded a stable, high- Q mode. Mesh scal of 1.4 was finalized based on convergence plots, and a ring radius of $19.562\mu m$ was decided on the basis of the second parametric sweep. The final characteristics for the $m = 354$ WGM are extracted and analyzed below.

4.1 Final Resonator Characteristics

4.1.1 Quality Factor (Q)

$$Q = \frac{\text{Re}(f)}{2 \cdot \text{Im}(f)} = \mathbf{167,058}$$

This Q_{rad} value is high, indicating that for a $20\mu m$ radius, the bending loss (radiation) is relatively low.

4.1.2 Effective Mode Volume (V_{eff})

The effective mode volume is calculated in post-processing by integrating the electric field energy density over the simulation domain and normalizing by its maximum value:

$$V_{eff} = \frac{\int \epsilon(\mathbf{r})|\mathbf{E}(\mathbf{r})|^2 dV}{\max[\epsilon(\mathbf{r})|\mathbf{E}(\mathbf{r})|^2]} = \frac{W_{stored,total}}{u_{max}}$$

- **Total Stored Energy ($W_{stored,total}$):** This value ($1.456 \times 10^{-31} Jm^3$) was obtained by defining a domain integration operator (e.g., ‘intop_domain’). This operator evaluates the volume integral of the electric field energy density, $W_e = \frac{1}{2}\mathbf{D} \cdot \mathbf{E}^* = \frac{1}{2}\epsilon_0\epsilon_r|\mathbf{E}|^2$, over the entire simulation domain. The COMSOL Wave Optics module provides this as a predefined variable.
- **Peak Energy Density (u_{max}):** This value ($1.409 \times 10^{-15} J$) was found using a maximum operator (e.g., ‘maxop_domain’ or a ‘Surface Maximum’ operator on the 2D domain). This operator finds the peak value of the same ‘wop.We’ variable anywhere in the computational domain: $u_{max} = \max(\text{wop.We})$.

This yields an effective mode volume of:

$$\mathbf{V_{eff} = \frac{1.4555 \times 10^{-31} J}{1.4093 \times 10^{-15} J/m^3} = 1.0328 \times 10^{-16} m^3}$$

4.2 Calculated Purcell Factor (F_p)

With Q and V_{eff} determined, we can now calculate the maximum achievable Purcell factor using the formula derived in Section 2.2:

$$F_p = \frac{3}{4\pi^2} \left(\frac{\lambda}{n} \right)^3 \left(\frac{Q}{V_{eff}} \right)$$

Substituting Q and V_{eff} in the fomula:

$$F_p \approx \mathbf{2.27}$$

This simulation predicts a maximum ZPL enhancement of $F_p \approx 2.27$, assuming ideal emitter placement and orientation.

4.3 Discussion and Literature Comparison

The primary figure of merit for this application is F_p , which is directly proportional to the ratio Q/V_{eff} . Our analysis reveals the fundamental limitation of the MRR geometry for this specific application.

- **On Q-Factor:** The simulated $Q_{rad} = 1.67 \times 10^5$ is a respectable value for a visible-light resonator. However, it is an ideal upper bound. State-of-the-art experimental SiN microdisks (a similar WGM geometry) have demonstrated Q-factors as high as 3.4×10^6 at $\lambda = 660$ nm.[10] Furthermore, any fabrication process will introduce sidewall roughness, creating a dominant scattering loss channel (Q_{scat}) that will likely reduce the final experimental Q to a value well below the simulated Q_{rad} .
- **On Mode Volume:** The V_{eff} is the critical flaw. This value is enormous. This is not a design error, but an inherent property of a WGM. To achieve a high Q_{rad} , bending loss must be minimized, which requires a large radius R . Since the mode is delocalized over the entire circumference $L = 2\pi R$, the mode volume ($V_{eff} \approx A_{mode} \cdot L$) scales directly with R .
- **On Purcell Factor:** The resulting $F_p \approx 2.27$ is modest. It is comparable to early experimental demonstrations (e.g., $F_p = 2.0$ reported in) but falls far short of the $F_p \sim 30$ (ZPL-only enhancement) or $F_p \sim 50$ [9] required to create a truly efficient and deterministic spin-photon interface.

The core of the issue is the Q/V_{eff} trade-off.

- **Microring Resonators (MRRs)** are a *high-Q, high- V_{eff}* geometry. They excel at applications requiring long photon lifetimes (e.g., filters, sensors, nonlinear optics) but are poor at maximizing F_p .

- **Photonic Crystal (PhC) Cavities** are the opposite. They are a *low- V_{eff}* , (*typically*) *lower- Q* geometry. They are designed to confine light to a single-point defect, achieving $V_{eff} \sim 1(\lambda/n)^3$ or even smaller.[4]

Because V_{eff} is in the denominator of the F_p equation, a 3000-fold reduction in V_{eff} is far more impactful than a 10-fold increase in Q .

This is clearly illustrated by comparing this work to state-of-the-art (SotA) literature values in Table 1.

Table 1: Comparison of simulated MRR performance with state-of-the-art (SotA) cavity-NV/SiV systems in the visible spectrum.

Device Platform	Q-Factor	$V_{eff} [(\lambda/n)^3]$	Purcell Factor (F_p)
This Work (SiN MRR, Sim.)	1.67×10^5	~ 29.869	2.27 (Calculated)
SotA SiN Microdisk (Exp.) [10]	3.4×10^6	High (Unspecified)	N/A
SotA SiN PhC (Sim.) [4]	1.4×10^6	0.78	> 1000 (Projected)
SotA Diamond PhC (Exp.)	1.6×10^5	Low (Unspecified)	18 (Demonstrated, SiV)
SotA Hybrid SiN (Exp.)	N/A	N/A	2.0 (Overall), ~ 30 (ZPL only)

The comparison in Table 1 is stark. The Q -factor of this work is in all the cases 10 times lower than the experimentally demonstrated high-performance PhCs. However, the V_{eff} is over 40x larger than what is achievable in a SiN PhC. This is the reason why the MRR yields a trivial $F_p \approx 2.27$, while PhC designs project $F_p > 1000$ and have experimentally demonstrated $F_p \sim 18 - 30$.

5 Conclusion and Future Outlook

5.1 Summary of Findings

This report has documented the systematic design and computational analysis of a SiN microring resonator intended to enhance the 637 nm ZPL of a diamond NV center. A 2D axisymmetric eigenfrequency model in COMSOL Multiphysics was validated through a mesh convergence study and optimized via a ring radius sweep.

The final design (a 250x500 nm waveguide in a $R = 19.562\mu m$ ring) was found to support an $m = 354$ WGM with a radiation-limited Q-factor of $Q = 167,058$ and an effective mode volume of $V_{eff} = 1.033 \times 10^{-16} \text{ m}^3$. This volume, when normalized, is exceptionally large at $V_{eff} \approx 29.869$.

5.2 Key Conclusion

The key result of this analysis is the calculation of a modest theoretical Purcell factor of $F_p \approx 2.27$. This work serves as a clear numerical demonstration of the fundamental Q/V_{eff} trade-off. While the microring resonator platform is highly effective at achieving a high Q-factor (by minimizing radiation loss through a large radius), its intrinsically large mode volume makes it a fundamentally unsuitable geometry for maximizing the Purcell factor. The Q/V_{eff} ratio is orders of magnitude lower than what is required for a high-efficiency spin-photon interface.

References

- [1] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, "The nitrogen-vacancy colour centre in diamond," *Physics Reports*, vol. 528, no. 1, pp. 1–45, 2013.
- [2] L. Childress and R. Hanson, "Diamond NV centers for quantum computing and quantum sensing," *MRS Bulletin*, vol. 38, no. 2, pp. 134–138, 2013.
- [3] A. Faraon, C. Santori, Z. Huang, V. M. Acosta, and R. G. Beausoleil, "Coupling of nitrogen-vacancy centers in diamond to a silicon nitride photonic crystal cavity," *Physical Review Letters*, vol. 109, no. 3, p. 033604, 2012.
- [4] V. M. Acosta, E. Bauch, M. P. Ledbetter, A. Waxman, L.-S. Bouchard, and D. Budker, "Temperature dependence of the nitrogen-vacancy magnetic resonance in diamond," *Physical Review Letters*, vol. 104, no. 7, p. 070801, 2010.
- [5] E. M. Purcell, "Spontaneous emission probabilities at radio frequencies," *Physical Review*, vol. 69, p. 681, 1946.
- [6] S. L. Mouradian, N. H. Wan, T. T. Tran, and D. R. Englund, "Rectangular Photonic Crystal Nanobeam Cavities in Bulk Diamond," *Applied Physics Letters*, vol. 111, no. 2, p. 021103, 2017.
- [7] M. W. Doherty, N. B. Manson, P. Delaney, and L. C. L. Hollenberg, "The negatively charged nitrogen-vacancy centre in diamond: the electronic structure," *New Journal of Physics*, vol. 13, no. 2, p. 025019, 2011.
- [8] P. Rabiei, W. H. P. Pernice, C. Schuck, and H. X. Tang, "High-Q silicon nitride ring resonators for integrated photonics at 637 nm," *Applied Physics Letters*, vol. 102, no. 17, p. 171106, 2013.
- [9] K. Luke, A. L. Gaeta, and M. Lipson, "Broadband mid-infrared frequency comb generation in a Si₃N₄ microresonator," *Optics Express*, vol. 23, no. 21, pp. 27717–27723, 2015.
- [10] D. Englund, H. A. L. F. P. M. F. A. F. R. G. B. J. V., "Deterministic coupling of a single nitrogen-vacancy center to a photonic crystal cavity," *Nano Letters*, vol. 10, no. 10, pp. 3922–3926, 2010.